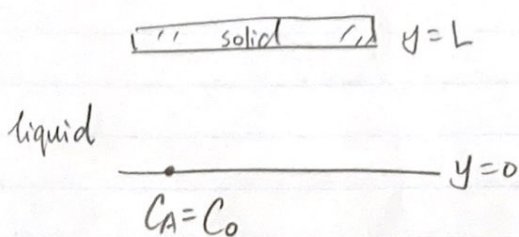


Q1)



$A \rightarrow B$   
 $A \rightarrow C$   
 $A \rightarrow F$

Find

 $C_A$ 

$$0 = -\frac{\partial \dot{N}_A y}{\partial y} + \dot{N}_{A, \text{net gen}}$$

$$0 = -\frac{d}{dy} (-D_{\text{eff}} \frac{dC_A}{dy}) + \dot{N}_{A, \text{net gen}}$$

$$\dot{N}_{A, \text{net gen}} = -k_B - k_C - (k_F \cdot C_A)$$

$$\frac{d}{dy} \left( \frac{dC_A}{dy} \right) = \frac{k_B + k_C + k_F \cdot C_A}{D_{\text{eff}}}$$

$$\frac{d^2 C_A}{dy^2} = \frac{k_B + k_C}{D_{\text{eff}}} + \frac{k_F}{D_{\text{eff}}} \cdot C_A$$

$$\frac{d^2 C_A}{dy^2} - \frac{k_F}{D_{\text{eff}}} C_A - \left( \frac{k_B}{D_{\text{eff}}} + \frac{k_C}{D_{\text{eff}}} \right) = 0 \quad m^2 = \frac{k_F}{D_{\text{eff}}}$$

$$C_A = C_1 e^{my} + C_2 e^{-my} + \left( \frac{k_B + k_C}{D_{\text{eff}}} \cdot \frac{1}{m^2} \right)$$

Bc: at  $y=0$ ,  $C_A = C_0$ 

$$C_0 = C_1 + C_2 + \frac{k_B + k_C}{D_{\text{eff}}} \cdot \frac{D_{\text{eff}}}{k_F}$$

$$C_0 = C_1 + C_2 + \frac{k_B + k_C}{k_F}$$

$$C_1 = C_0 - C_2 - \frac{k_B + k_C}{k_F}$$

at  $y=L$ ,  $-D_{\text{eff}} \frac{dC_A}{dy} = 0$ 

$$\frac{dC_A}{dy} = m C_1 e^{my} - m C_2 e^{-my}$$

$$-D_{\text{eff}} \cdot \left( \sqrt{\frac{k_F}{D_{\text{eff}}}} C_1 e^{mL} - \sqrt{\frac{k_F}{D_{\text{eff}}}} C_2 e^{-mL} \right) = 0$$

$$-\sqrt{KF} \cdot \sqrt{D_{eff}} C_1 e^{mL} + \sqrt{KF} \sqrt{D_{eff}} C_2 e^{-mL} = 0$$

$$-\sqrt{KF} \sqrt{D_{eff}} \left( C_0 - C_2 - \frac{k_B + k_C}{KF} \right) + \sqrt{KF} \sqrt{D_{eff}} C_2 e^{-mL} = 0$$

$$-\sqrt{KF} \sqrt{D_{eff}} C_0 + C_2 \sqrt{KF} \sqrt{D_{eff}} + \frac{k_B + k_C}{\sqrt{KF}} + \sqrt{KF} \sqrt{D_{eff}} C_2 e^{-mL} = 0$$

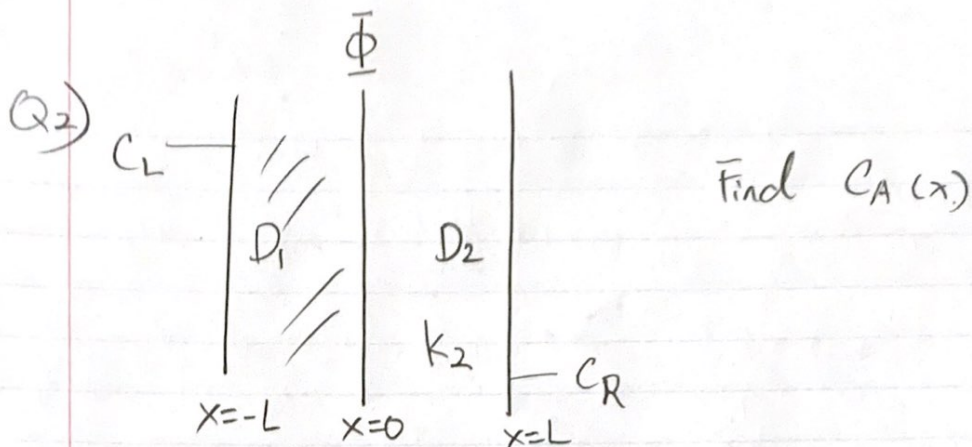
$$C_2 = \frac{\sqrt{KF} \sqrt{D_{eff}} \cdot C_0 - \frac{k_B + k_C}{\sqrt{KF}}}{\underbrace{(\sqrt{KF} \sqrt{D_{eff}} + e^{-mL})}_Q}$$

$$C_A = \left( C_0 - Q - \frac{k_B + k_C}{KF} \right) e^{my} + Q e^{-my} + \frac{k_B + k_C}{D_{eff}} \cdot \frac{D_{eff}}{KF}$$

Bonus:  $\frac{dn_A}{dt} = D_{eff} \frac{d}{dy} \frac{dC_A}{dy}$

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m-1:

$$D_1 \frac{d^2 C_{A,1}}{dx^2} = 0 \quad -L < x < 0$$

$$C_{A,1}(x) = Ax + B$$

m-2:

$$D_2 \frac{d^2 C_{A,2}}{dx^2} + (-k_2) = 0$$

$$\frac{d^2 C_{A,2}}{dx^2} = \frac{k_2}{D_2}$$

$$\frac{dC_{A,2}}{dx} = \frac{k_2}{D_2} x + E$$

$$C_{A,2} = \frac{k_2}{D_2} \frac{x^2}{2} + Ex + F$$

Bc: at  $x = -L$   $C_{A,1}(-L) = C_L$   
 $C_L = -AL + B$

at  $x = L$   $C_{A,2}(L) = \frac{k_2}{D_2} \frac{L^2}{2} + EL + F = C_R$

Bc: at  $x = 0$   $\bar{\Phi} = \frac{C_{A,1}(0)}{C_{A,2}(0)} = \frac{B}{F}$

Bc: at  $x = 0$   $-D_1 \frac{dC_{A,1}}{dx} = -D_2 \frac{dC_{A,2}}{dx}$

$$B = F \cdot \Phi$$

$$\frac{dC_{A,1}}{dx} = A$$

$$\frac{dC_{A,2}}{dx} = \frac{k_2}{D_2} x + E$$

$$-D_1(A) = -D_2(E)$$

$$C_L = -AL + B$$

$$A = \frac{D_2}{D_1} \cdot E$$

$$C_R = \frac{k_2}{D_2} \frac{L^2}{2} + EL + F$$

$$E = \frac{C_R - \frac{k_2}{D_2} \frac{L^2}{2} - F}{L}$$

$$C_L = - \left( \frac{D_2}{D_1} \cdot \left( \frac{C_R - \frac{k_2}{D_2} \frac{L^2}{2} - F}{L} \right) (L) \right) + (F \Phi)$$

$$C_L = - \frac{D_2}{D_1} (C_R - \frac{k_2}{D_2} \frac{L^2}{2} - F) + F \Phi$$

$$C_L = - \frac{D_2}{D_1} C_R + \frac{k_2}{D_1} \frac{L^2}{2} + \frac{F D_2}{D_1} + F \Phi$$

$$F \left( \frac{D_2}{D_1} + \Phi \right) = C_L + \frac{D_2}{D_1} C_R - \frac{k_2}{D_1} \frac{L^2}{2}$$

$$F = \frac{C_L + \frac{D_2}{D_1} C_R - \frac{k_2}{D_1} \frac{L^2}{2}}{\frac{D_2}{D_1} + \Phi}$$

$$B = F \cdot \Phi = \frac{C_L + \frac{D_2}{D_1} C_R - \frac{k_2}{D_1} \frac{L^2}{2}}{\frac{D_2}{D_1} + \Phi} \cdot \Phi$$

$$E = \frac{-C_R - \frac{k_2}{D_2} \frac{L^2}{2} - C_L - \frac{D_2}{D_1} C_R + \frac{k_2}{D_1} \frac{L^2}{2}}{L}$$

$$A = \frac{D_2}{D_1} \cdot \left( \frac{-C_R - \frac{k_2}{D_2} \frac{L^2}{2} - C_L - \frac{D_2}{D_1} C_R + \frac{k_2}{D_1} \frac{L^2}{2}}{L} \right)$$

$$\Rightarrow C_{A,1}(x) = Ax + B$$

$$C_{A,2}(x) = \frac{k_2}{D_2} \frac{x^2}{2} + Ex + F$$



$$\frac{dC_1}{dr} = A.$$

$$N_A''(x) = -D_1 \frac{dC_1}{dr} = -D_1 A$$

$$N_A''(x) = -D_{\text{eff}} \cdot \frac{dC}{dx}$$

$$D_{\text{eff}} = \frac{D_2}{D_1} \left( -\frac{k_2}{D_2} \frac{L^2}{2} - \left(1 + \frac{D_2}{D_1}\right) + \frac{k_2}{D_1} \frac{L^2}{2} \right)$$

$$N_A''(x) = -D_{\text{eff}} \frac{dC}{dx} \leftrightarrow \text{either } C_{A,1} \text{ or } C_{A,2}$$

